Let's try it out! We submit a question to the chatbot using `chain.run()` and then view the response.

```
question = {"query": "Will AI impact work forces in the US ? "}
answer = chain.run(question)
```

```
 Yes, AI, including LLMs, is projected to have a significant impact on the US workforce. According to a study, approximatel
y 19% of jobs have at least 50% of their tasks exposed to LLMs when considering both current model capabilities and anticip
ated LLM-powered software. The impact is expected to be more pronounced in higher-wage occupations and could have various e
ffects such as augmenting or displacing human labor, affecting job quality, and influencing inequality. However, social, ec
onomic, regulatory, and other factors will also play a role in determining the actual outcomes. Further research is necessa
ry to understand the broader implications of AI advancements on the workforce.
```

Figure 6.22 – Testing our chatbot with chain.run()

Once our chatbot model is ready, we track and log our model with MLflow. We do this for the tracking and version control but also because there is a direct integration with the MLflow LangChain flavor that will help us seamlessly serve this model using Databricks Model Serving, and then later connect to hosted applications outside of the Databricks platform.

```
from mlflow.models import infer_signature
from mlflow.tracking.client import MlflowClient

mlflow.set_registry_uri("databricks-uc")
model_name = f"{catalog}.{database_name}.mlaction_chatbot_model"

with mlflow.start_run(run_name="mlaction_chatbot_rag") as run:
    signature = infer_signature(question, answer)
    model_info = mlflow.langchain.log_model(
        chain,
        loader_fn=get_retriever,
        artifact_path="chain",
        registered_model_name=model_name,
        pip_requirements=[
            "mlflow==" + mlflow.__version__,
            "langchain==" + langchain.__version__,
            "databricks-vectorsearch",
        ],
        input_example=question,
        signature=signature
    )
    #------------------------
    import mlflow.models.utils
    mlflow.models.utils.add_libraries_to_model(
        f"models:/{model_name}/{get_latest_model_version(model_name)}"
    )
```

Figure 6.23 – Tracking and logging our model